# Lecture 5: Fisher Information and Interval Estimators

IEOR E4709: Statistic Analysis and Time Series
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## 1 Fisher Information

Let $L(\theta|x)$ denote the likelihood function parameterized by $\theta$. The *Fisher information* associated with the model is the matrix given by

$$I(\theta) := \mathbb{E}_\theta \left[ \nabla_\theta \log L(\theta|X) \nabla_\theta \log L(\theta|X)^\top \right], \tag{1}$$

where $\nabla_\theta f(\theta) = (f_{\theta_1}(\theta), f_{\theta_2}(\theta), \ldots, f_{\theta_d}(\theta))$ is the (column vector) gradient of $f$ with respect to $\theta$. Moreover, $\mathbb{E}_\theta$ denotes the expectation taken with respect to the probability density function $L(\theta|x)$ (fixing $\theta$ and viewed as a function of $x$, the likelihood function is a probability density function). Intuitively, the Fisher information captures the variability of the gradient of the log-likelihood function (i.e., of $\nabla_\theta \log L(\theta|X)$). The function $l(\theta|x) := \nabla_\theta \log L(\theta|X)$ is also called the *score function*.

In a family of distributions for which the score function has high variability, we intuitively expect the estimation of the parameter $\theta$ to be easier. This is because different $\theta$'s will produce changes in value for $l(\theta|x)$.

Let us make a few useful observations. First, we notice that the score function has mean zero under $\mathbb{P}_\theta$: we have

$$\begin{aligned} \mathbb{E}_\theta \left[ l(\theta|X) \right] &= \int L(\theta|x) l(\theta|x) dx = \int \frac{\nabla_\theta L(\theta|x)}{L(\theta|x)} L(\theta|x) dx \\ &= \int \nabla_\theta L(\theta|x) dx = \nabla_\theta \int L(\theta|x) dx = \nabla_\theta 1 = 0, \end{aligned}$$

where we have assumed that the integral and differential operator can be exchanged (dominated convergence theorem applies).

Using this result, we can provide an alternative representation of the Fisher information matrix. Indeed, we know that

$$0 = \mathbb{E}_\theta \left[ l(\theta|X) \right] = \int l(\theta|x) L(\theta|x) dx \tag{2}$$

Differentiating with respect to $\theta$ on both sides of equation (2) gives (recall that $\theta$ is thought of as

a column vector)

$$\begin{aligned} 0 &= \int \frac{\partial^2 \log L(\theta|x)}{\partial\theta\partial\theta^T} L(\theta\big|x)dx + \int l(\theta|x)\left(\nabla_\theta L(\theta|x)\right)^\top dx \\ &= \mathbb{E}_\theta\left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top}\right] + \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right], \end{aligned} \tag{3}$$

where the $(j,k)$-th entry of $\left[\frac{\partial^2 \log L(\theta|x)}{\partial\theta\partial\theta^T}\right]$ is $\frac{\partial^2 \log L(\theta|x)}{\partial\theta_j\partial\theta_k}$. The second equality above follows from the fact that

$$\begin{aligned} \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right] &= \int l(\theta|x)\nabla_\theta \log L(\theta|x)^\top L(\theta|x)dx \\ &= \int l(\theta|x)\frac{1}{L(\theta|x)}\nabla_\theta L(\theta|x)^\top L(\theta|x)dx \\ &= \int l(\theta|x)\nabla_\theta L(\theta|x)^\top dx \end{aligned}$$

Recall that $\mathbb{E}_\theta\left[l(\theta|X)\right] = 0$. Hence,

$$Var_\theta\left(l(\theta|X)\right) = \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right] - \mathbb{E}_\theta\left[l(\theta|X)\right]\mathbb{E}_\theta\left[l(\theta|X)^\top\right] = \mathbb{E}_\theta\left[l(\theta|X)l(\theta|X)^\top\right]$$

Consequently, recalling the definition of $I(\theta)$ in Eq. (1), it follows from Eq. (3) that

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top}\right] = Var_\theta(l(\theta|X))$$

Hence, the $(j,k)$-th entry is $\left[I(\theta)\right]_{j,k} = -\mathbb{E}\left[\frac{\partial^2 \log L(\theta|x)}{\partial\theta_j\partial\theta_k}\right]$.

**Remark 1.1.** *The alternative representation of the Fisher Information matrix given above relies heavily on the fact that the score function has mean zero, which in turn relies crucially on the exchanges of differentiation and expectation rigorous. For this to hold, we must have some conditions on the probability density functions we consider. One simple condition sufficient to make this work is provided by the Lebesgue's dominated convergence theorem. Let $f : X \times \Theta \to \mathbb{R}$ be a differentiable function. For a fixed base measure $\mu$, assume there exists a function $g$ such that $g(x) \geq ||\nabla_\theta f(x,\theta)||$ for all $\theta$, where $\int g(x)d\mu(x) < \infty$. Then, in this case, we have $\nabla_\theta \int f(x,\theta)d\mu(x) = \int \nabla_\theta f(x,\theta)d\mu(x)$ by the mean-value theorem and the definition of a derivative.*

The above representation has the intuitive property that if we have $n$ i.i.d observations coming from the model $L(\theta|x)$, then the information content grows linearly because $\log L(\theta\big|(X_1, X_2, \ldots, X_n)) = \sum_{i=1}^n \log L(\theta|X_i)$.

**Example:** We provide an example to show the close connection of Fisher information to the level of estimation accuracy. Suppose we want to estimate the mean of a Bernolli($p$) random variable, where $p \in [0,1]$, from a sample of i.i.d. Bernoulli random variables $X_1, X_2, \ldots, X_n$. First, notice that $\mathbb{P}(X_i = x) = p^x(1-p)^{1-x}$, where $x \in \{0,1\}$. Then $\nabla_p \log(\mathbb{P}(X_i = x)) = \frac{x}{p} - \frac{1-x}{1-p}$. Therefore, the Fisher information associated with an individual observation $X_i$ is given by

$$I(p) = \mathbb{E}_p\left[(\nabla_p \log L(p\big|X_i))^2\right] = \mathbb{E}_p\left[\left(\frac{X_i}{p} - \frac{1-X_i}{1-p}\right)^2\right] = \mathbb{E}_p\left[\frac{(X_i-p)^2}{(p(1-p))^2}\right] = \frac{1}{p(1-p)}.$$

The sample mean $\bar{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased estimator, whose mean squared error is given by

$$\mathbb{E}\left[(\bar{p}-p)^2\right] = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n}p(1-p) = \frac{1}{nI(p)},$$

where $I(p)$ is the Fisher information for a single Bernoulli($p$) observation (the denominator $nI(p)$ is the Fisher information associated with the data set $x_1, x_2, \ldots, x_n$). It also states that the more informative an observation is (higher $I(p)$), and the more the variance of an unbiased estimator is reduced. This is an instance of a more general result: the Fisher information provides a lower bound on the mean squared error of any unbiased estimator, i.e., the variance of any unbiased estimator cannot be smaller than the inverse of the Fisher Information matrix.

## 2 Cramer-Rao Lower Bound

In this section and going forward, we will use $I(\theta)$ to denote the Fisher information of the entire data set of observations associated with the parameter $\theta$.

**Proposition 2.1** (**Cramer-Rao lower bound).** *Suppose $\theta$ is a d-dimensional vector, and let $T(X)$ be an unbiased estimator of $\theta$, i.e., $\mathbb{E}[T(X)] = \theta$. Then*

$$Cov(T(X)) \geq I(\theta)^{-1},$$

*where a matrix $A \geq B$ means that $A - B$ is positive semidefinite, i.e. the smallest eigenvalue of this matrix is nonnegative. If $\theta$ is a scalar, then it simplifies to $Var(T(X)) \geq \frac{1}{I(\theta)}$*

Above, the $(j,k)$-th entry of the covariance matrix is $[Cov(T(X))]_{jk} = Cov((T(X))_j, (T(X))_k)$.

**Example:** Let us compute the variance-covariance matrix of the sample mean estimator based on a sequence of $n$ i.i.d. Gaussian random variables $X_1, X_2, \ldots, X_n$. We want to verify whether it attains the Cramer-Rao lower bound. Recall that the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased estimator of $\mu$, where $\mu$ is the mean of the Gaussian random variable $X_i$. We also know that $Var(\bar{X}) = \frac{\sigma^2}{n}$. Recall that the log-likelihood function of a Gaussian is given by

$$\log L(\mu|x) = -n\log\left(\sqrt{2\pi}\right) - \frac{n}{2}\log\left(\sigma^2\right) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$$

Hence,

$$I(\mu) = -\mathbb{E}_\mu\left[\frac{\partial^2 \log L(\mu|X)}{\partial^2 \mu}\right] = \frac{n}{\sigma^2},$$

where $X = (X_1, X_2, \ldots, X_n)$. Hence, $Var(\bar{X}) = \frac{1}{I(\mu)}$. This means that the sample mean is the best estimator of $\mu$, i.e., among all those unbiased estimator of $\mu$ it is one that achieves the CRLB.